Field Failure and Warranty Analysis

Cab mounting and suspension bracketry, FY24
Including the reversal of the aluminium cab mounting bracket

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Prepared by	Field Quality and Reliability
Reviewed by	Chief Engineer - Chassis Systems; Chair, VAVE Committee
Population	41,280 vehicles, FY22 to FY24 build
Claims analysed	821 within the bracketry family
Distribution	VAVE Committee; Product Engineering; Service Engineering

Summary

Eight hundred and twenty-one warranty claims against cab mounting and suspension bracketry were analysed for FY24. Mount-face fatigue cracking is the dominant mode at 38 per cent of claims and is concentrated almost entirely in the aluminium cab mounting bracket introduced in FY22. That change has been reversed. This report records the failure mechanism, the reversal decision and the conditions under which a future aluminium proposal could be reconsidered, so that the reversal is not repeated as a new idea.

Contents

1	Population and method	2
2	Claims by failure mode	2
3	Mount-face fatigue cracking	3
4	Reversal of the aluminium cab mounting bracket	4
5	Other modes	5
6	Conditions for reconsideration	5

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1 Population and method

The analysed population is 41,280 vehicles built between April 2021 and March 2024 across the HCV rigid, tipper and tractor-trailer programmes. Claims were extracted from the warranty system for the bracketry family and were individually reviewed; 94 claims were reclassified and 37 were excluded as accident damage. Returned parts were available for 212 claims and were examined at the materials laboratory.

1.1 Exposure normalisation

Claim rates are expressed per thousand vehicles at twenty-four months in service, so that build years with different exposure can be compared. Tipper vehicles accumulate load cycles considerably faster than rigid vehicles at the same age, and tipper rates are shown separately wherever the mode is duty-sensitive.

2 Claims by failure mode

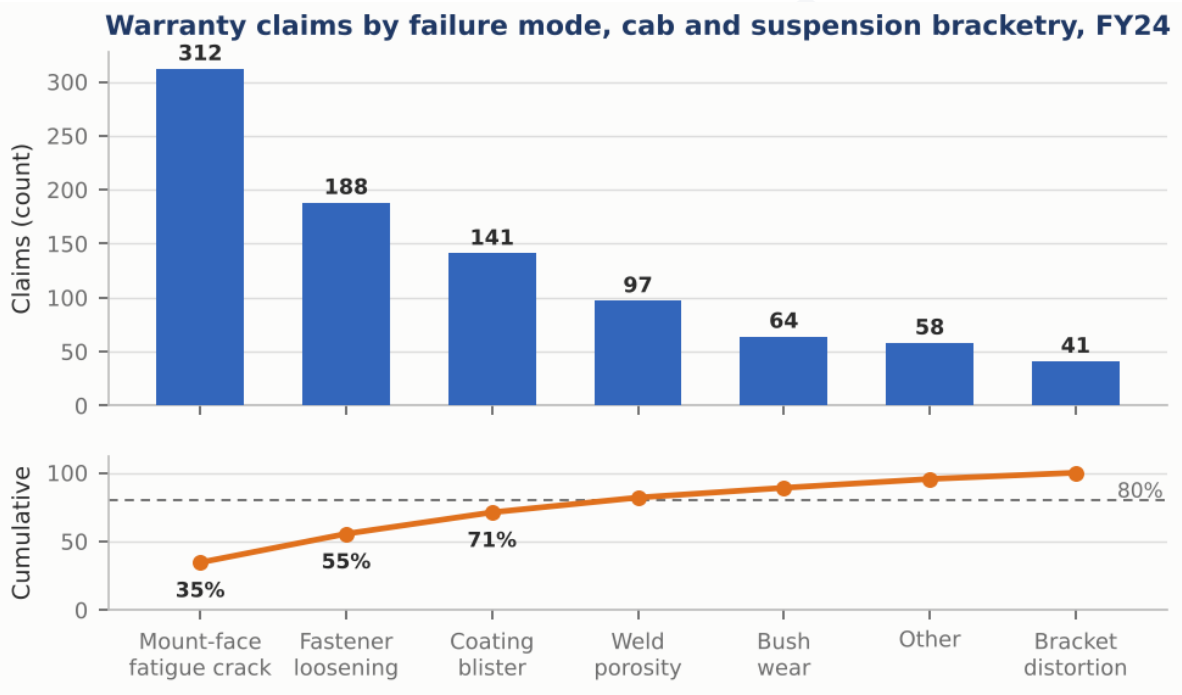


Figure 1 - Warranty claims by failure mode for cab and suspension bracketry, FY24. The upper panel shows claim counts; the lower panel shows the cumulative share. Three modes account for 71 per cent of claims.

Failure mode	Claims	Rate / 1000 @ 24 mo	Tipper rate	Trend
Mount-face fatigue crack	312	7.56	14.21	Falling since reversal
Fastener loosening	188	4.55	6.02	Stable
Coating blister	141	3.42	3.61	Falling
Weld porosity	97	2.35	2.48	Stable
Bush wear	64	1.55	2.94	Stable
Bracket distortion	41	0.99	2.11	Rising
Other	58	1.41	1.66	-

3 Mount-face fatigue cracking

3.1 Mechanism

Cracking initiates at the machined mount face, at the edge of the clamped area on the inboard side, and propagates across the section. Laboratory examination of 68 returned brackets found beach marks consistent with high-cycle fatigue in all cases, with no evidence of casting porosity, inclusion or overload. The initiation site coincides with the edge of the washer footprint.

The mechanism is fretting-initiated fatigue at the clamped interface. Under cab roll, the mount face experiences a small cyclic relative displacement against the steel washer. In the steel bracket this is benign. In the aluminium bracket the lower hardness permits fretting damage to establish, and the resulting surface condition acts as a fatigue initiation site.

Finding 3.1. The failure is an interface problem, not a bulk strength problem. The aluminium bracket meets its static and fatigue targets as a component. It fails in service because of the behaviour of the clamped joint between two dissimilar materials, which the component-level validation did not represent.

3.2 Why validation did not predict it

The FY21 validation applied a load spectrum to the bracket through a rigid fixture with a hardened interface plate. That fixture reproduced the bracket loads correctly and did not reproduce the clamped joint. Fretting at the mount face was therefore absent from the test. The bracket achieved 1.34 times target life on the bench and reached 62 per cent of target life in the field.

Finding 3.2. A bench result of 1.34 times target and a field result of 0.62 times target is not test scatter. It indicates that the test did not contain the failure mechanism. Validation fixtures for clamped joints in dissimilar materials must reproduce the joint, not just the load.

3.3 Distribution by programme

Programme	Population	Claims	Rate / 1000 @ 24 mo
HCV rigid	22,140	118	5.33
HCV tipper	9,860	140	14.21
Tractor-trailer	6,420	41	6.39
RMC and COWL	2,860	13	4.55

The tipper rate is 2.7 times the rigid rate. Tipper duty combines higher cab roll amplitude with a higher cycle count, and the two effects compound. Any future proposal to carry an aluminium bracket to the tipper programme must be assessed against the tipper duty cycle specifically.

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4 Reversal of the aluminium cab mounting bracket

4.1 History

Date	Event	Reference
Mar 2021	Idea submitted: steel to aluminium cab mounting bracket	VC-21-0188
Aug 2021	Component validation passed at 1.34 x target life	VAL/CAB/2021/044
Feb 2022	Released to production, all HCV rigid programmes	ECN 22-0319
Nov 2022	First field claims received, rigid programme	-
Jun 2023	Claim rate exceeds action threshold on tipper	FQ/ALT/2023/08
Jan 2024	Root cause confirmed as fretting-initiated fatigue	MAT-LAB/2024/011
Mar 2024	Reversal to steel approved	VC-24-0019, ECN 24-0588
Sep 2024	Reversal complete across all programmes	-

4.2 Consequence

Measure	Value
Mass benefit forgone, per vehicle	2.84 kg
Cost benefit forgone, per vehicle	INR 214
Warranty cost incurred, FY23 to FY24	INR 31.4 million
Vehicles in field with the aluminium bracket	18,420
Field campaign status	Monitored, no recall
Tooling written off	INR 8.6 million

Finding 4.1. The reversal is recorded in the idea register as VC-24-0019, a separate entry from the original VC-21-0188. A search that finds the original returns an implemented change with no indication that it was withdrawn. Seven re-submissions of the aluminium bracket proposal were received in FY24 H1 alone.

4.3 What was NOT the cause

The following were examined and excluded. They are recorded because each has been proposed as an explanation and each would lead to a different and incorrect corrective action.

- Material grade. AA6061-T6 to specification in all 68 examined parts; hardness and conductivity within range.
- Casting quality. No porosity or inclusion at or near any initiation site.
- Bolt torque. Residual torque measured on 44 returned joints; 41 within specification. Loosening is a separate mode, section 5.1.
- Section thickness. Increasing section was modelled and moves the initiation site without extending life, because the mechanism is at the surface.

5 Other modes

5.1 Fastener loosening

One hundred and eighty-eight claims, second most frequent. Concentrated on joints using a plain washer under a flanged nut without a prevailing-torque feature. The corrective action released in FY24 Q2 standardises the chassis bracket joint on a flanged bolt with a serrated face; see CVOEM/VAVE/ES-STD/2023/004 section 4.

5.2 Coating blister

One hundred and forty-one claims, falling. Associated with 8 um clear zinc plating on exposed chassis brackets, which does not meet the 720 hour requirement. Affected part numbers have been moved to zinc-nickel. No further action required.

5.3 Bracket distortion

Forty-one claims and the only mode with a rising trend. Concentrated on the 5 mm crossmember with local doubler released in FY24. Distortion occurs at the doubler edge rather than in the thinned section. Under investigation; provisional finding is a weld sequence effect rather than a design deficiency.

Finding 5.1. The rising distortion trend is associated with a recent mass-reduction change. It is recorded here so that the FY24 crossmember change is not assessed as an unqualified success in future benchmarking.

6 Conditions for reconsideration

The aluminium cab mounting bracket proposal is closed, not prohibited. It may be reconsidered if all of the following are satisfied. This list exists so that a future proposal can be assessed against stated conditions rather than being refused by reference to an outcome.

- The clamped joint is redesigned to eliminate fretting at the mount face, for example by a hardened steel insert or a bonded interface, and the redesign is described in the proposal.
- Validation is performed on a fixture that reproduces the clamped joint in dissimilar materials, not a rigid interface plate.
- Validation includes the tipper duty cycle, not the rigid duty cycle alone.
- The proposal states the mass and cost benefit at current material rates per CVOEM/VAVE/MC-IX/2025/003, not at the FY21 rates used in the original case.
- The proposal explicitly references VC-21-0188 and VC-24-0019 and states what has changed since the reversal.

Recommendation. Any submission proposing an aluminium cab mounting bracket that does not address all five conditions should be closed by citing this section, without a fresh engineering assessment.

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